

Timing Inaccessibility and the Projection Bound: Resolving Maxwell’s Demon for Continuous Biological Substrates

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April 3, 2026

Abstract

We show that the thermodynamic advantage of biological, continuous substrates over digital simulators arises from *timing inaccessibility*: below the Landauer threshold, temporal order cannot be irreversibly registered without dissipating $\geq k_{\text{B}}T \ln 2$ per binary decision. Consequently, exponentially many micro-trajectories map to the same observable outcome (*path degeneracy*). Continuous high-dimensional substrates exploit this by integrating sub-Landauer couplings during evolution and paying only at *projection* (dimensional collapse to a low-dimensional output).

We derive a **Projection Bound** for quasistatic projection at effective temperature T_{eff} :

$$E_{\text{collapse}} \geq k_{\text{B}}T_{\text{eff}} \left(\ln \frac{N_{\varepsilon, \text{pre}}}{N_{\varepsilon, \text{post}}} - \text{KL}(p^{\text{pre}} \parallel U_{\text{pre}}) + \text{KL}(p^{\text{post}} \parallel U_{\text{post}}) \right) \quad (0.1)$$

which reduces under typical-set conditions to $E_{\text{collapse}} \geq k_{\text{B}} T_{\text{eff}} \ln(N_{\varepsilon, \text{pre}}/N_{\varepsilon, \text{post}})$. Combined with a **Temporal Registration Bound** (order over M bins needs $\log_2 M!$ bits), we quantify the gap: enumerative digital tracking scales exponentially with dimension, whereas projection cost scales like $\ln G \sim D$.

For biologically plausible parameters, we estimate degeneracies of 10^{42} – 10^{94} (protein folding) and 10^{50} – 10^{100} (neural dynamics) as upper bounds under independence assumptions. The framework reconciles stochastic resonance (amplitude detection) with order inaccessibility, and clarifies why analog/neuromorphic systems gain efficiency by deferring projection.

Keywords: Maxwell’s demon, timing inaccessibility, path degeneracy, Landauer principle, dimensional collapse

Significance

What’s new? A geometric thermodynamic bound showing that continuous high-dimensional substrates pay once at projection, not per micro-decision.

Why it matters? It explains biology’s energy efficiency without violating the second law: sense weakly (sub-Landauer), integrate across many modes, and collapse late.

Numbers: Order-of-magnitude path degeneracies of 10^{42} – 10^{94} (proteins) and 10^{50} – 10^{100} (neural) arise for typical $(D_{\text{eff}}, \tau_c, \Delta t)$ (see Appendix A for parameter sensitivity).

Testable: (i) $\tau_c \uparrow \rightarrow \text{accuracy} \uparrow$; (ii) D_{eff} shifts across frequency-specific constraint channels with task; (iii) decision-time heat $\approx k_{\text{B}} T_{\text{eff}} \ln(N_{\text{pre}}/N_{\text{post}})$ at attojoule scale (~ 0.03 – 1 aJ). For example, ordering $M=10$ bins requires $\log_2(10!) \cdot k_{\text{B}} T \ln 2 \approx 6.3 \times 10^{-20}$ J (0.063 aJ) at 300 K, representing a lower-end example within this range.

1 Introduction

High-dimensional systems exist as physical and mathematical objects. When you need to compute what such a system does, you face an implementation choice: simulate it on a low-dimensional digital substrate (bit strings), or instantiate it physically as a high-dimensional system. This paper quantifies the thermodynamic efficiency gap between these approaches.

1.1 The Biological Maxwell’s Demon

Living systems operate in a Maxwell–demon–like mode. They:

- create local order (cells, organisms, structured neural activity) from stochastic environments,
- extract work from thermal and chemical gradients,
- maintain far-from-equilibrium organization over decades, and
- process information to guide adaptive responses.

The energetic puzzle is stark: a human brain performs sophisticated computations at ~ 20 W, whereas comparable digital systems require orders of magnitude more power [?, ?]. If biological systems *acquired and recorded* information as discrete bits, Landauer’s principle would demand a dissipation of at least $k_B T \ln 2$ per bit. Yet measured neural energetics are far below what such pervasive bit–level bookkeeping would imply.

We consider three possibilities:

1. **Physics violation** — biology evades the second law (ruled out empirically).
2. **Hidden costs** — the energy is dissipated elsewhere, unmeasured (possible but unparsimonious).

3. **Different accounting** — continuous high-dimensional substrates store and use information as *structural correlations*, not as stabilized bits; the thermodynamic payment appears only at *projection* (this paper’s thesis).

We demonstrate (iii): biological systems maintain information as correlations in high- D coherence fields and defer irreversible registration. The cost is paid when those correlations are collapsed to discrete actions — the dimensional projection. Between acquisition and projection, information exists physically (in coupled dynamics) without bit erasure, enabling apparent sub-Landauer operation locally while preserving global thermodynamic consistency.

Central mechanism (preview). Below the Landauer threshold ($k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J at 300 K), *temporal order cannot be irreversibly registered* without dissipating at least $k_B T \ln 2$ per binary timing decision. This *timing inaccessibility* creates exponential path degeneracy: many micro-trajectories map to the same observable outcome. Digital systems that track order pay per recorded bit; continuous high- D substrates integrate weak, sub-Landauer couplings and pay once at projection. See ??? for the Temporal Registration Bound and the Projection Bound.

Critical distinction: Throughout this paper, “sub-Landauer detection” refers to reversible coupling and integration that does not leave a persistent record. Energy costs enter when order or state is *stabilized as memory* (logically irreversible projection). This distinction resolves apparent paradoxes: biological systems can sense and integrate weak signals while deferring the thermodynamic cost of recording.

Scope note. “Maxwell’s demon” here is a metaphor for correlation-enabled work extraction under correct thermodynamic accounting; no violation of the second law is implied.

Main Results

Temporal Registration Bound (TRB): Irreversibly recording the total order of M temporal bins requires $\log_2(M!)$ bits (by Stirling: $\log_2(M!) \geq M \log_2 M - M \log_2 e$ for large M), hence:

$$E_{\min}^{\text{time}} \geq k_B T \ln 2 \cdot \log_2(M!) \quad (1.1)$$

Projection Bound (PB): Quasistatic projection from $N_{\epsilon, \text{pre}}$ to $N_{\epsilon, \text{post}}$ distinguishable states dissipates:

$$E_{\text{collapse}} \geq k_B T_{\text{eff}} \ln(N_{\epsilon}(D_{\text{eff}})/N_{\epsilon}(D')) \quad (1.2)$$

under typical-set approximation (equiprobable cells).

Path Degeneracy Scaling: With coherence time τ_c and temporal resolution Δt :

$$\Omega(\Delta t) \sim \exp \left[\frac{D_{\text{eff}}(\Delta t)}{\kappa} \ln \frac{\tau_c}{\Delta t} \right] \quad (1.3)$$

where $\kappa \geq 1$ accounts for correlations.

1.2 Efficiency Gap: Where the Energy Goes

Digital algorithm tracking the path:

- To uniquely index among G options: $\log_2 G$ bits required
- Protein folding: $\log_2(10^{48-100}) \approx 160-332$ bits
- Energy dissipated during tracking: $E_{\text{digital}} \gtrsim (160-332)k_B T \ln 2 \approx 4.6 \times 10^{-19} - 9.5 \times 10^{-19}$ J (ideal Landauer floor; practical CMOS is $10^4-10^6 \times$ higher, while fully reversible logic can approach the floor only with severe speed and noise-tolerance tradeoffs [?, ?])
- This applies even if history is later discarded—the cost was paid during recording

Continuous substrate (deferred projection):

- System samples all $\sim 10^{48}$ – 10^{100} conformations via thermal fluctuations
- No irreversible records made during exploration
- Pay only at projection: $E_{\text{cont}} \geq k_{\text{B}}T_{\text{eff}} \ln 2$ for binary collapse ("native" vs. "misfolded")

Efficiency gain: ~ 160 – $330\times$ at theoretical Landauer limit. Contemporary CMOS operates at $\sim 10^4$ – $10^6\times$ Landauer limit, raising practical gap to $\sim 10^5$ – $10^8\times$ [?, ?]. While adiabatic reversible computing could theoretically narrow this gap [?, ?], practical implementations face severe speed and noise-tolerance tradeoffs. Digital architectures pay per recorded bit; continuous substrates defer recording and pay at projection. Analog/neuromorphic systems likewise gain efficiency by deferring discretization to the projection boundary.

1.3 Notation and Conventions

1.4 Roadmap

Section 2 formalizes timing inaccessibility. Section 3 derives the Projection Bound. Section 4 quantifies path degeneracy. Section 5 gives a continuous (Kuramoto) example. Section 6 relates to prior work and discusses implications including camera-engine duality. Appendix A provides parameter sensitivity analysis for degeneracy estimates. Appendix B describes biological implementations across scales. Appendix C provides methods for estimating D_{eff} and κ from data.

Table 1: Key symbols and definitions

Symbol	Definition
$k_{\text{B}}T \ln 2$	Landauer bound per bit ($\approx 2.87 \times 10^{-21}$ J at 300 K)
T	Bath temperature (K); used in TRB
$T_{\text{eff}}(\omega)$	Effective temperature at frequency ω via FDT (K); defined operationally at collapse mode; used in PB
D_{eff}	Effective dimensionality, $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$
D'	Post-collapse effective dimensionality (output manifold dimension)
N_ε	Covering number: count of ε -balls covering manifold
G, Ω	Path degeneracy (number of micro-trajectories per macro-outcome); used synonymously
τ_c	Coherence time, $\int_0^\infty \langle r(t)r(0) \rangle dt$ (s)
Δt	Temporal resolution (s)
κ	Correlation factor ≥ 1 (reduces degeneracy from independence)
ε	Coarse-graining resolution
L	System size (same units as ε ; L/ε is dimensionless)
ξ	Correlation (coherence) length in capacity expressions
Conventions:	
Natural logarithms (nats) unless stated; $\log_2 x = \ln x / \ln 2 \approx 1.443 \ln x$	
Energy: $k_{\text{B}}T$ per nat or $k_{\text{B}}T \ln 2$ per bit	
Typical set: high-probability subset under source measure (Cover & Thomas)	

2 Timing Inaccessibility: The Physical Mechanism

2.1 Why Temporal Order Costs More Than Amplitude

Following Todd (2025) [?], measurement at energy E_{meas} extracts information bounded by:

$$I_{\text{max}} \lesssim \frac{E_{\text{meas}}}{k_{\text{B}}T \ln 2} \quad (2.1)$$

consistent with the AWGN limit where reliable bit recovery demands $E_b/N_0 \geq \ln 2$ [?, ?].

Detection capacity (AWGN, $E_b/N_0 \geq \ln 2$) is separate from recording/erasure cost (Landauer, $\geq k_{\text{B}}T \ln 2$ per bit). The former bounds channel capacity (cf. Verdú's $E_b/N_0 \geq \ln 2$); the latter applies when information is irreversibly stabilized as memory (Landauer's $k_{\text{B}}T \ln 2$).

When $E_{\text{meas}} \ll k_{\text{B}}T \ln 2$, you cannot extract even one bit reliably. This means timing

resolution fails.

Lemma 2.1 (Temporal Registration Bound). *To irreversibly register the temporal order of M bins requires at least $H \geq \log_2(M!)$ bits. By Landauer’s principle, the minimal dissipated energy is:*

$$E_{\min}^{\text{time}} \geq k_{\text{B}}T \ln 2 \cdot \log_2(M!) \quad (2.2)$$

For partial order distinguishing equivalence classes of sizes m_j :

$$E_{\min}^{\text{time}} \geq k_{\text{B}}T \ln 2 \cdot \log_2 \left(\frac{M!}{\prod_j m_j!} \right) \quad (2.3)$$

At sub-Landauer energies where $E_{\text{meas}} \ll k_{\text{B}}T \ln 2$, the number of binary timing decisions that can be irreversibly registered approaches zero.

Proof sketch. There are $M!$ possible total orders of M bins. Uniquely specifying one order requires $H = \log_2(M!)$ bits (by Stirling: $\log_2(M!) \approx M \log_2 M - M \log_2 e$). Irreversibly writing these bits dissipates at least $k_{\text{B}}T \ln 2$ per bit by Landauer’s principle [?]. $\square$

Clarification: A reversible comparator that leaves no persistent record can, in principle, avoid dissipation. The energetic bound applies when the order is *stabilized as memory* (logically irreversible).

Concrete example: At $T = 300$ K, ordering $M = 10$ bins minimally dissipates $\log_2(10!) \cdot k_{\text{B}}T \ln 2 \approx 21.8 \times 2.87 \times 10^{-21}$ J $\approx 6.3 \times 10^{-20}$ J.

2.2 Stochastic Resonance: Why It Detects Amplitude But Not Order

Natural objection: If stochastic resonance enables detection of sub-Landauer signals through population integration [?, ?], doesn’t this eliminate path degeneracy?

Answer: No. Stochastic resonance provides:

- Amplitude/presence detection ($\text{SNR} \sim \sqrt{N}$ scaling)
- Population-level integration across N coupled units
- Knowledge that coordination occurred

It *cannot* provide:

- Timing of individual events (which fluctuation occurred when)
- Temporal ordering (did event A precede B?)
- Which specific coordination pattern was realized

For $M = 10$ temporal bins, distinguishing all orderings requires $\log_2(10!) \approx 21.8$ bits, costing $\geq 6.3 \times 10^{-20}$ J. If individual signal energies are $E \sim 10^{-23}$ J (sub-Landauer), measurement energy exceeds signal energy by $6000\times$. Population integration can detect *that* 10 events occurred, but not *when* or in what order.

2.3 High Dimensionality: The Key to Sub-Landauer Sensing

The puzzle: If individual events are sub-Landauer, how does the system sense anything?

The solution: The physical substrate is formally infinite-dimensional with power-law mode distribution. At any finite energy/temporal resolution, D_{eff} is finite and grows with precision. At energy threshold E_{thresh} , a finite number of modes $D_{\text{eff}}(E_{\text{thresh}})$ carry energy $\geq E_{\text{thresh}}$.

With D_{eff} accessible modes, a signal of energy $E_{\text{signal}} \ll k_B T \ln 2$ per component can still be detected if integrated:

$$E_{\text{total}} \approx N_{\text{eff}} E_{\text{signal}}, \quad N_{\text{eff}} \propto D_{\text{eff}} \times \ln \frac{\tau_c}{\Delta t} \quad (2.4)$$

Even if $E_{\text{signal}} \ll k_B T \ln 2$ per channel, $E_{\text{total}} \gtrsim k_B T \ln 2$ achieves detectability.

Example: Neural population with $D_{\text{eff}} \sim 1000$ accessible modes can detect signals where each mode receives only $E_{\text{signal}} \sim 10^{-3} k_B T \ln 2$ because population integration yields effective energy $E_{\text{total}} \sim 1000 \times 10^{-3} k_B T \ln 2 = k_B T \ln 2$, meeting the detection threshold.

Ephaptic coupling: Extracellular fields $\sim 0.1\text{--}1$ mV/mm over neuronal membrane ($C_m \sim 1$ $\mu\text{F}/\text{cm}^2$) yield work $\sim 10^{-23}\text{--}10^{-22}$ J per neuron ($\approx 3 \times 10^{-3}\text{--}3 \times 10^{-2}$ times $k_B T \ln 2$) (see [?] for field strengths and scaling).

2.4 Why Digital Systems Cannot Do This

Fundamental architectural difference:

Digital systems:

- Binary bit requires $\geq k_B T \ln 2$ to be irreversibly registered
- Must receive supra-Landauer signals per input bit
- Cannot integrate sub-Landauer signals across channels without first amplifying each channel above threshold (costs energy)
- Must discretize each input above Landauer threshold before processing

Continuous high-D systems:

- Can couple weakly (sub-Landauer per link) to many sources
- Integrate via population dynamics
- Collapse only at output
- Pay for dimensional reduction, not per-interaction bit registration

The degeneracy is not a bug—it’s what enables weak coupling to be computationally useful. Path degeneracy and sensing capability are coupled: exponentially many microstates allow sensitivity to distributed sub-Landauer perturbations.

3 The Projection Bound

3.1 What Are We Paying For?

When forcing a high-dimensional system to produce a low-dimensional output, you must dissipate heat. The cost has three components:

1. Geometric squashing ($\ln N_{\varepsilon,\text{pre}}/N_{\varepsilon,\text{post}}$): How many configurations are you destroying? For continuous manifolds, $N_{\varepsilon} \sim (L/\varepsilon)^{D_{\text{eff}}}$ where L/ε is dimensionless.

2. Pre-collapse non-uniformity ($-\text{KL}(p^{\text{pre}}\|U_{\text{pre}})$): Were you using all states uniformly? If the system had a heavy-tailed distribution, you were effectively more constrained than geometry suggests. This *lowers* the bound (you've already "paid" some entropy reduction).

3. Post-collapse specificity ($+\text{KL}(p^{\text{post}}\|U_{\text{post}})$): How peaked is your final state? This *increases* the bound—you're creating more order.

Typical-set approximation: For high-dimensional systems with many weakly-interacting modes, equipartition and maximum entropy drive toward near-uniform occupancy. When both pre- and post-collapse distributions are approximately uniform, KL terms vanish and you get the clean geometric bound.

3.2 Effective Temperature: Operational Definition

For systems out of equilibrium, define T_{eff} operationally via the fluctuation-dissipation relation [?]:

$$S_{\eta}(\omega) = 2k_{\text{B}}T_{\text{eff}}(\omega) \text{Re}[\Gamma(\omega)] \quad (3.1)$$

where $S_{\eta}(\omega)$ is noise power spectral density, $\text{Re}[\Gamma(\omega)]$ is the dissipative component of linear response, and $\Gamma(\omega)$ describes how the system responds to weak perturbations: $\langle x(\omega) \rangle = \Gamma(\omega)F(\omega)$.

Measurement protocol:

1. Measure noise spectrum $S_\eta(\omega)$
2. Apply weak sinusoidal perturbation at ω_0 (e.g., inject 50–200 pA membrane-equivalent drive in cortical slice)
3. Measure phase-resolved response to extract $\text{Re}[\Gamma(\omega_0)]$
4. Ratio gives $T_{\text{eff}}(\omega_0) = S_\eta(\omega_0)/(2k_B \text{Re}[\Gamma(\omega_0)])$ —an observable, not a fitted parameter

In equilibrium: $T_{\text{eff}} = T$. Out of equilibrium: $T_{\text{eff}}(\omega)$ is operational, evaluated at collapse mode ω_0 . Note that in active biological systems driven by ATP hydrolysis or other energy sources, T_{eff} can exceed ambient temperature T [?], reflecting enhanced effective noise from non-thermal fluctuations.

3.3 Main Theorem

Theorem 3.1 (Projection Bound for Continuous Systems). *Let $\{C_i\}_{i=1}^{N_\varepsilon}$ be an ε -coarse graining of the accessible manifold and $p^{\text{pre}}, p^{\text{post}}$ be the pre/post distributions over cells. For a quasistatic projection that reduces support from $N_{\varepsilon, \text{pre}}$ to $N_{\varepsilon, \text{post}}$, the mean dissipated heat at effective temperature T_{eff} (evaluated at the dominant collapse mode ω_0) satisfies:*

$$\boxed{E_{\text{collapse}} \geq k_B T_{\text{eff}} \left(\ln \frac{N_{\varepsilon, \text{pre}}}{N_{\varepsilon, \text{post}}} - \text{KL}(p^{\text{pre}} \parallel U_{\text{pre}}) + \text{KL}(p^{\text{post}} \parallel U_{\text{post}}) \right)} \quad (3.2)$$

where $U_{\text{pre/post}}$ are uniform on their supports. (This reproduces (??) under the typical-set approximation.)

Under typical-set approximation (equiprobable cells):

$$E_{\text{collapse}} \geq k_B T_{\text{eff}} \ln(N_\varepsilon(D_{\text{eff}})/N_\varepsilon(D')) \quad (3.3)$$

With power-law covering, $N_\varepsilon \sim (L/\varepsilon)^D$ for fixed L/ε :

$$E_{\text{collapse}} \geq k_B T_{\text{eff}} (D_{\text{eff}} - D') \ln(L/\varepsilon) \quad (3.4)$$

Proof outline. **Step 1:** Define coarse-grained entropy as $S_\varepsilon = k_B(\ln N_\varepsilon - \text{KL}(p\|U))$ where U is uniform over the N_ε cells.

Step 2: The projection reduces support and alters p :

$$\Delta S_{\text{sys}} = S_\varepsilon^{\text{post}} - S_\varepsilon^{\text{pre}} \quad (3.5)$$

$$= k_B[\ln N_{\varepsilon,\text{post}} - \ln N_{\varepsilon,\text{pre}} - \text{KL}(p^{\text{post}}\|U_{\text{post}}) + \text{KL}(p^{\text{pre}}\|U_{\text{pre}})] \quad (3.6)$$

Step 3: By the second law: $\Delta S_{\text{env}} \geq -\Delta S_{\text{sys}}$.

Step 4: Integrating heat at T_{eff} gives $E_{\text{collapse}} = T_{\text{eff}}\Delta S_{\text{env}} \geq -T_{\text{eff}}\Delta S_{\text{sys}}$, yielding ?? $\square$

Classical limit: Binary system with $N_\varepsilon = 2 \rightarrow 1$ recovers $k_B T \ln 2$.

Finite-time correction: Finite-time protocols add thermodynamic-length term $\mathcal{L}^2/(4\tau)$ [?]. For neural decisions at $\tau \sim \text{ms}$ timescales with typical $\mathcal{L} \sim 10$, excess dissipation $\sim 10^{-20}$ J is small compared to quasistatic bound ($\sim 10^{-19}$ – 10^{-18} J for $D_{\text{eff}} \sim 50$ – 200).

4 Path Degeneracy: Quantifying the Inaccessible

4.1 Definition and Scaling

At coarse-graining ε over window with temporal resolution Δt , $\Omega(\Delta t)$ is the number of resolution-distinguishable micro-trajectories consistent with the same macro-outcome.

For continuous manifold $\mathcal{M}$, the metric ε -entropy [?]:

$$H_\varepsilon = \ln N_\varepsilon(\mathcal{M}), \quad N_\varepsilon \sim \left(\frac{L}{\varepsilon}\right)^{D_{\text{eff}}} \quad (4.1)$$

with L/ε dimensionless.

As ε decreases, N_ε grows exponentially. Systems with large D_{eff} exhibit exponentially more distinguishable configurations.

For oscillator system with coherence time τ_c and temporal resolution Δt , the time-bandwidth product $\tau_c/\Delta t$ bounds independent temporal modes [?], yielding:

$$\Omega(\Delta t) \sim \exp \left[\frac{D_{\text{eff}}(\Delta t)}{\kappa} \ln \frac{\tau_c}{\Delta t} \right] \quad (4.2)$$

where $\kappa \geq 1$ accounts for correlations.

4.2 Neural-Scale Example

Consider neural dynamics with:

- Coherence time: $\tau_c \sim 100$ ms
- Coarse temporal resolution: $\Delta t_{\text{coarse}} = 20$ ms
- Fine temporal resolution: $\Delta t_{\text{fine}} = 0.1$ ms
- Dimensionality increase: $\Delta D_{\text{eff}} \approx 50$ (consistent with [?])
- Correlation redundancy: $\kappa \sim 2$ (estimated from phase-shuffle surrogates)

Path degeneracy ratio:

$$\begin{aligned} \text{Ratio} &\sim \exp \left[\frac{\Delta D_{\text{eff}}}{\kappa} \ln \left(\frac{\tau_c/\Delta t_{\text{fine}}}{\tau_c/\Delta t_{\text{coarse}}} \right) \right] \\ &= \exp \left[\frac{50}{2} \times \ln(200) \right] \approx 10^{58} \end{aligned} \quad (4.3)$$

Interpretation: One observable spike train at coarse resolution (20 ms bins) corresponds to $\sim 10^{58}$ distinguishable micro-trajectories at fine resolution (0.1 ms bins). This falls within the 10^{50} – 10^{100} range cited in the abstract (see Appendix A for full parameter sensitivity showing how variations in D_{eff} , κ , and temporal resolution produce this range).

5 Example: Kuramoto Oscillators

Consider N coupled phase oscillators:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) + \eta_i(t) \quad (5.1)$$

where $\langle \eta_i(t) \eta_j(t') \rangle = 2D \delta_{ij} \delta(t - t')$. State space is the N -torus $\mathcal{T}^N$.

Order parameter: $re^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}$

At angular resolution $\Delta\theta$, one macroscopic configuration (r, Ψ) corresponds to:

$$N_{\Delta\theta} \sim \left(\frac{2\pi}{\Delta\theta} \right)^{D_{\text{eff}}} \quad (5.2)$$

distinguishable microstates.

Effective dimensionality:

- Synchronized regime (strong coupling, $r \approx 1$): $D_{\text{eff}} \approx 2$ (Ott-Antonsen manifold for Lorentzian $g(\omega)$ under analyticity assumptions [?, ?])
- Incoherent regime (weak coupling, $r \approx 0$): $D_{\text{eff}} \approx N$
- Intermediate coupling: $2 < D_{\text{eff}} < N$

When the system collapses from exploring the full D_{eff} -dimensional manifold to a measurement outcome:

$$E_{\text{collapse}} \geq k_B T_{\text{eff}} (D_{\text{eff}} - D') \ln(2\pi/\Delta\theta) \quad (5.3)$$

This is the continuous analog of Landauer's bound—fully geometric, no discrete bits. Recent computational approaches encounter numerical failures ("covariance explosion") precisely where temporal resolution demands exceed thermodynamic accessibility [?]
—computational evidence for the bounds derived here.

6 Relation to Prior Work and Implications

6.1 Information-Theoretic Demons

Prior continuous demons [?, ?, ?] quantify costs via mutual information. Our contribution: (i) classical biological substrate at 300K, (ii) metric-entropy covering numbers at finite resolution, (iii) timing inaccessibility as mechanism, (iv) path degeneracy from dimensional structure.

Bennett [?] established that logical reversibility bounds energy only if computation is quasi-static and low-noise. Still et al. [?] showed predictive information confers thermodynamic advantage, but creating/storing it has minimal work cost. Kolchinsky & Wolpert [?] demonstrated that minimal work depends on changes in non-equilibrium distributions—aligning with our Projection Bound via $\ln N_\epsilon$ and KL terms.

6.2 Quantum Demons

Recent quantum demons [?, ?] show coherence-entropy trade-offs with mathematical isomorphism $S(\rho) \leftrightarrow H_\epsilon$. Physical mechanisms differ—quantum superposition (mK, μs) vs. classical path degeneracy (300K, ms)—yet both yield isomorphic bounds. The dimensional geometry may unify quantum and classical regimes, differing in mechanism (coherent superposition vs thermal degeneracy) but not fundamental structure.

6.3 Spatial Scales and Information Capacity

Voronel [?] and Bormashenko [?] established that biological informational capacity is constrained by spatial scales. Biological systems exploit *both* spatial and temporal fine-graining. Total capacity:

$$E_{\text{collapse}} \geq k_B T_{\text{eff}} (D_{\text{spatial}} + D_{\text{temporal}}) \ln(L/\xi) \quad (6.1)$$

6.4 Camera-Engine Duality: A New Demon Mechanism

These prior frameworks quantify costs but don't fully capture how biological substrates defer projection. This motivates the camera-engine duality below, which reframes the demon mechanism physically.

The classical demon is straightforward: a computer with memory that it periodically erases, paying $k_B T \ln 2$ per bit. The demon *is* the memory.

The biological demon is more subtle. It's not a memory device—it's a high-dimensional dynamical system that simultaneously functions as both **camera** (sensing environmental complexity) and **engine** (steering environmental dynamics). This dual role parallels both MacKenzie's observation that economic models both describe and shape markets [?] and Boyd's OODA loop framework [?], where adaptive systems continuously observe, orient, decide, and act upon their environment. Here the duality is simultaneous rather than sequential, with thermodynamic cost concentrated at the decision/collapse point.

Camera–Engine Duality

What the demon IS: A formally infinite-dimensional substrate with accessible slice D_{eff} at measurement threshold. Maintains dimensional isomorphism between internal and environmental structure.

What the demon DOES:

- 1. Camera (sensing):** Couples weakly (sub-Landauer, $E_{\text{link}} \ll k_{\text{B}}T \ln 2$) to exponentially many environmental modes. High D_{eff} enables integration: $E_{\text{total}} = \sum_i E_{\text{link},i} \gtrsim k_{\text{B}}T \ln 2$. Internal dynamics become correlated with environment—a structural map, not written bits.
- 2. Engine (steering):** Uses internal map to bias environmental dynamics via coordinated weak back-coupling across D_{eff} channels.
- 3. Collapse (payment):** When output crystallizes (protein folds, neuron spikes), D_{eff} collapses to discrete state, paying $\gtrsim k_{\text{B}}T_{\text{eff}} \ln(N_{\text{pre}}/N_{\text{post}})$.

Why weak coupling avoids immediate costs: Sub-Landauer coupling creates mutual information through physical correlation (stored in joint state), not epistemic records. Following Sagawa-Ueda [?], thermodynamic cost appears only when correlation is measured and stabilized as discrete memory—the dimensional collapse.

Resolution of the Paradox The demon doesn't "extract information" in the sense of writing bits. It *becomes isomorphic* to environmental structure via dimensional resonance. Information exists in the *correlation* between internal and external structure—physical (coupling energy) not epistemic (written records). Thermodynamic cost appears only when correlation must collapse to discrete action.

High-D systems can maintain mutual information through structural isomorphism without irreversible bit registration. A protein folding in response to chaperone binding doesn't "read" the chaperone's state—the protein's 10^{48} microstates resonantly couple to chaperone conformations. The mutual information exists in physical correlation, not written memory. Only when committing to native fold does dimensional collapse occur.

Path degeneracy is the signature that structural correlation exists without bit registration. The system effectively explores a vast ensemble of micro-trajectories via thermal fluctuations while deferring any persistent record of order. Because individual coupling energies are $\ll k_B T \ln 2$, you cannot track "which path when" without measurement energy exceeding signal energy (back-action). Digital systems require supra-Landauer inputs per bit and cannot exploit this integration regime.

6.5 The Accuracy-Efficiency Tradeoff

What digital computers buy with their energy cost:

- Bit-exact computation: deterministic and reproducible
- Known state: query any variable, get exact value
- Auditable history: complete, verifiable trace
- Precision arithmetic: arbitrary precision (hardware-limited)
- Logical guarantees: formal verification, proof of correctness

What high-D physical substrates sacrifice for efficiency:

- Path indeterminacy: cannot know which of 10^{48} paths taken
- Thermal noise: inherent randomness from $\sim k_B T$ fluctuations
- Approximate outputs: "protein folded" not "path #37,594,284,719"
- Non-reproducibility: exact micro-trajectory differs each time

Digital architectures pay per recorded bit to guarantee precision and auditability. Continuous substrates defer recording and gain energy efficiency for order-insensitive tasks. Neither is "better"—they solve different problems. Biology rarely needs bit-exact computation; it needs robust pattern recognition and approximate optimization.

6.6 Computational Irreducibility

When timing is thermodynamically inaccessible, predictive compression is lower-bounded by emulation [?, ?, ?]. Temporal order is not thermodynamically accessible as a persistent record until irreversible registration. The temporal structure is *created* by registration, not revealed. Any predictor must perform the same registration operations to create that structure—therefore must emulate the system up to collapse points.

This isn't a computational limitation but fundamental physics. The "when" you're trying to predict doesn't exist until the projection creates it through dimensional collapse. In Igamberdiev's framework [?], biological organization involves anticipatory systems where constraints precede values. Camera-engine duality enables this: the system maintains correlations (relational structure) without computing trajectories, allowing emergent functional meaning at collapse.

6.7 Limitations and Caveats

Theoretical:

- Bounds stated at fixed ε for quasistatic protocols
- Typical-set reduction is conservative; KL terms tighten bounds for non-uniform distributions
- Sub-Landauer energy estimates are order-of-magnitude; detailed simulations remain desirable
- Future work could simulate finite-time corrections using thermodynamic geometry [?]

Task-dependent:

- Degeneracy numbers represent upper bounds assuming statistical independence
- Correlations reduce effective Ω by factors $\kappa \geq 1$ (see Appendix A for sensitivity)

- Efficiency claims strongest for order-insensitive tasks where outputs depend on integrated statistics
- For timing-critical tasks (e.g., precise spike-timing codes), path degeneracy and advantages diminish

Common objections:

Objection 1: "Digital doesn't have to store full history—you can compress or discard intermediate states."

Reply: Compression reduces storage but not the record-selection cost at the moment of committing to one trajectory class. Deciding among G order classes and stabilizing that decision as persistent, queryable record still incurs $\geq k_B T \ln G$ dissipation.

Objection 2: "Analog front-ends (e.g., in neuromorphic chips) can also integrate weak sub-Landauer signals before digitization."

Reply: Exactly our point. The energy cost appears at the analog-to-digital boundary or wherever memory is formed—the projection from continuous high-D dynamics to discrete records. Analog systems that defer this projection gain the efficiency we quantify.

Objection 3: "Is path degeneracy physical or merely epistemic?"

Reply: Both, but the physical aspect is primary. At any fixed ε , the count of thermodynamically distinguishable states is objective. Below ε , distinctions require measurement energy exceeding signal energy, making finer structure thermodynamically inaccessible (back-action), not merely unknown.

7 Conclusion

Maxwell's demon paradox extends to continuous substrates via timing inaccessibility and dimensional collapse. Below Landauer threshold, temporal ordering becomes thermodynamically inaccessible—you cannot irreversibly register "when" without spending $\geq k_B T \ln 2$ per

binary timing decision. This creates path degeneracy: exponentially many micro-trajectories map to the same observable outcome.

The biological demon uses high-dimensional substrates to efficiently compute high-dimensional dynamics via camera-engine duality: sensing environmental complexity via weak coupling, steering it via coordinated weak back-coupling, maintaining dimensional isomorphism without writing bits. Information exists in structural correlation, not epistemic records. Thermodynamic cost appears only at dimensional collapse.

Digital computers use low-dimensional substrates (bit strings) to simulate high-dimensional dynamics, explicitly tracking each coordinate and paying $k_B T \ln 2$ per bit. State-space and enumerative simulation cost grow exponentially with D , whereas minimal dissipation paid by a high-D substrate scales like $k_B T_{\text{eff}} \ln G \sim D$. The system explores all paths in parallel via thermal fluctuations using sub-Landauer coupling energies. Because individual interactions are $\ll k_B T \ln 2$, you cannot track "which path when" without measurement energy exceeding signal energy.

Neither substrate is "better"—they solve different problems. Digital computation pays the energy cost for deterministic, bit-exact results with complete epistemic access, essential for tasks requiring precision, reproducibility, and verifiability. High-D physical substrates sacrifice this precision to achieve energy efficiency—you cannot know which path was taken, only that the robust macro-outcome emerged. Biology rarely needs bit-exact computation; it needs robust pattern recognition and approximate optimization.

To know what water does, use water. To know what a protein does, use a protein. To know what neurons do, use neurons. Digital simulation incurs exponential state-space costs for many such tasks, though reversible computing may asymptotically approach thermodynamic bounds with severe speed and noise-tolerance tradeoffs [?, ?].

The framework presented here—timing inaccessibility, path degeneracy, and dimensional projection—provides a quantitative foundation for understanding biological efficiency without invoking violations of fundamental physical law.

Acknowledgments

The author thanks reviewers for constructive feedback.

Declarations

Funding: None. **Competing interests:** None. **Generative AI use:** Claude 4.5 Sonnet (Anthropic) for drafting and conceptual development; GPT-5 (OpenAI) and Grok (xAI) for feedback and review. Author reviewed all content and takes full responsibility.

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A Parameter Sensitivity Analysis for Path Degeneracy

This appendix provides worked examples showing how parameter choices lead to the headline degeneracy ranges cited in the main text.

A.1 Protein Folding Example

Parameters:

- Micro-level states: $N_{\text{micro}} \sim 10^{48}\text{--}10^{100}$ (rotamer libraries to continuous torsions)
- Meso-level states: $N_{\text{meso}} \sim 10^6$ (folding intermediates)
- Temporal coherence: $\tau_c \sim 10^{-6}\text{--}1$ s (folding time)
- Fine resolution: $\Delta t_{\text{fine}} \sim 10^{-12}$ s (bond vibration)
- Coarse resolution: $\Delta t_{\text{coarse}} \sim 10^{-9}$ s (conformational transition)
- Effective dimensionality: $D_{\text{eff}} \sim 20\text{--}50$ (backbone degrees of freedom)
- Correlation factor: $\kappa \sim 5\text{--}10$ (folding funnel constraint)

Calculation:

$$\log_{10} \Omega = \frac{D_{\text{eff}}}{\kappa} \log_{10} \left(\frac{\tau_c}{\Delta t_{\text{fine}}} \right) + \log_{10} \left(\frac{N_{\text{micro}}}{N_{\text{meso}}} \right) \quad (\text{A.1})$$

$$\approx \frac{20-50}{5-10} \times \log_{10}(10^6-10^{12}) + \log_{10}(10^{42}-10^{94}) \quad (\text{A.2})$$

$$\approx 42-94 \quad (\text{A.3})$$

A.2 Neural Population Example

Parameters:

- Population size: $N \sim 1000$ neurons
- Coherence time: $\tau_c \sim 100$ ms
- Fine resolution: $\Delta t_{\text{fine}} \sim 0.1$ ms (spike timing precision)
- Coarse resolution: $\Delta t_{\text{coarse}} \sim 20$ ms (behavioral bins)
- Effective dimensionality: $D_{\text{eff}} \sim 50-200$ (from large-scale recordings)
- Correlation factor: $\kappa \sim 2-5$ (estimated as $D_{\text{eff}}^{\text{shuffle}}/D_{\text{eff}}^{\text{data}}$ under phase-shuffle surrogates)

Calculation:

$$\log_{10} \Omega = \frac{\Delta D_{\text{eff}}}{\kappa} \log_{10} \left(\frac{\tau_c/\Delta t_{\text{fine}}}{\tau_c/\Delta t_{\text{coarse}}} \right) \quad (\text{A.4})$$

$$= \frac{50-200}{2-5} \times \log_{10}(200) \quad (\text{A.5})$$

$$\approx 23-92 \quad (\text{A.6})$$

Note: To avoid double counting, we do not add a separate "voltage microstate" multiplier; the scaling in ?? already counts resolution-distinguishable micro-trajectories at the chosen $(\varepsilon, \Delta t)$.

A.3 Sensitivity Table

Table 2: $\log_{10} \Omega$ across parameter ranges for neural example

$(\Delta D_{\text{eff}}, \kappa)$	$\tau_c/\Delta t = 50$	$\tau_c/\Delta t = 200$	$\tau_c/\Delta t = 500$
(30, 2)	25	34	40
(50, 2)	42	58	68
(80, 2)	68	92	108
(50, 3)	28	38	45
(50, 5)	17	23	27

Key observations:

- Degeneracy remains exponentially large ($\Omega \sim 10^{17}-10^{108}$) across plausible ranges
- Correlations (higher κ) reduce $\log_{10} \Omega$ proportionally
- Time-bandwidth ratio has logarithmic effect, so moderate changes in $\tau_c/\Delta t$ produce manageable shifts
- Upper bounds assume independence; real biological systems have additional constraints

B Appendix B: Biological Implementations Across Scales

The timing inaccessibility framework applies from molecular to organismal scales wherever continuous dynamics operate below the Landauer threshold.

B.1 Molecular Scale: Protein Folding

Protein folding exemplifies path degeneracy at the molecular level. A typical protein explores $\sim 10^{48}-10^{100}$ conformational microstates en route to its native fold, yet folding time is only $\sim 10^{-6}-1$ s.

Energetics: Thermal energy $k_B T \approx 4.1 \times 10^{-21}$ J at 300 K; individual hydrogen bonds $\sim 2-10 k_B T$; hydrophobic contacts $\sim 1-5 k_B T$. While individual bond energies are supra-

Landauer, the energetic cost of irreversibly registering the timing and order of the full conformational trajectory is sub-Landauer per timing decision.

Path degeneracy: For a 100-residue protein with ~ 3 conformations per residue, microstate space $\sim 3^{100} \approx 10^{48}$ to 10^{100} , but coarse-grained intermediate states $\sim 10^3$ – 10^6 . Path degeneracy $G \sim 10^{42}$ – 10^{94} (upper bounds; folding funnels reduce effective degeneracy via correlation factor $\kappa \sim 5$ – 10).

Collapse: Thermodynamic cost appears at native state stabilization. Estimated $\Delta G_{\text{fold}} \sim 5$ – 15 kcal/mol $\approx (3.5 \times 10^{-20}$ – $1.0 \times 10^{-19})$ J, consistent with collapsing $D_{\text{eff}} \sim 20$ – 50 effective dimensions.

B.2 Neural Population as Camera-Engine

Consider a neural population making a perceptual decision. The substrate is formally infinite-dimensional (continuous electromagnetic fields, cross-frequency phase coupling, continuous membrane voltage), but at typical recording resolutions (1 ms, 100 μ V sensitivity), only $D_{\text{eff}} \sim 100$ is thermodynamically accessible.

Camera phase (0–100 ms): Environmental photon arrivals at $\sim 10^6$ rods/cones. Each photon at 500–600 nm has $E \sim (3.3$ – $4.0) \times 10^{-19}$ J (~ 80 – $96 k_B T$), but timing is sub-Landauer. Accessible modes $D_{\text{eff}} \sim 100$ integrate timing patterns. Internal voltage landscape becomes isomorphic to environmental light pattern through weak coupling. No bits written—structural correlation emerges.

Engine phase (concurrent): Attention modulates sensory gain via weak top-down connections. Each feedback synapse $\sim 10^{-22}$ J $\ll k_B T \ln 2$, but $D_{\text{eff}} \sim 100$ feedback channels coordinately bias processing. Environment steered toward task-relevant features using internal map built during camera phase.

Collapse (at decision, ~ 150 ms): Motor output: "left" vs "right" (binary). Dimensional collapse: accessible slice $D_{\text{eff}} \sim 100 \rightarrow D' = 1$. Dissipation: $\sim 100 k_B T \ln 2 \approx 3 \times 10^{-19}$ J. Digital would pay this during every timestep; biological pays once at decision.

B.3 Cross-Scale Pattern

Table 3: Path degeneracy across biological scales

System	Event energy	D_{eff}	Estimated $\log_{10} \Omega$
Protein folding	$1\text{--}10k_{\text{B}}T$	20–50	42–94
Ca^{2+} waves	$5\text{--}20k_{\text{B}}T$	30–100	25–40
Gene regulation	$5\text{--}15k_{\text{B}}T$	10–30	15–20
Neural (ephaptic)	$\sim 10^{-3}k_{\text{B}}T \ln 2$	50–200	50–100

Common mechanism: Thermal noise explores vast micro-trajectory spaces; weak fields bias ensembles; thermodynamic cost concentrates at collapse to discrete outputs.

C Appendix C: Estimating D_{eff} and κ from Data

Data and preprocessing. Acquire $X \in \mathbb{R}^{T \times M}$ (time $\times$ channels). Band-limit to the frequency band of interest with zero-phase FIR; extract analytic signal $z_m(t)$ via Hilbert transform for phase-based measures.

Feature matrix. Either (i) use covariance $C = \frac{1}{T-1}(X - \bar{X})^\top (X - \bar{X})$, or (ii) build a phase-coherence matrix W with entries $W_{ij} = \left| \frac{1}{T} \sum_t e^{i(\phi_i(t) - \phi_j(t))} \right|$ (inter-site phase coherence, ISPC [?]). Normalize W to unit trace to avoid scale artifacts.

Eigen-spectrum and participation ratio. Let $\{\lambda_k\}$ be eigenvalues of C (or W), sorted descending. Define

$$D_{\text{eff}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}.$$

Report $D_{\text{eff}}(t)$ in a sliding window (e.g., 200 ms with 10 ms hop) to track task-locked changes.

Correlation factor κ . Construct S surrogate datasets by phase-shuffling each channel: $z_m(t) \mapsto e^{i\theta_m} z_m(t)$ with random θ_m , preserving power spectra but destroying cross-channel

timing. Compute $D_{\text{eff}}^{(\text{sur})}$ on each surrogate; define

$$\kappa \equiv \frac{D_{\text{eff}}^{\text{data}}}{\text{median}_s D_{\text{eff}}^{(\text{sur})}} \quad (\kappa \geq 1).$$

Use κ in Ω -scaling as in ??.

Robustness notes. (1) Z-score channels before C to prevent amplitude-dominated modes. (2) Repeat analyses across bands (theta/alpha/beta), aligning with task epochs. (3) Control for sample size by fixing window length and applying Ledoit–Wolf shrinkage on C if M is large relative to T .